Continuous random variables

Antonio de Leon

Probability and Statistics for Engineers Fall 2026

Monday, October 26	Densities, CDFs, and expectation
Wednesday, October 28	Exponential and normal models
Friday, October 30	Joint densities and marginals

Course support and deadlines

Ed Discussion: questions, announcements, and private messages. No Canvas Inbox.

Open Ed from Canvas All times Pacific Full contact details and meeting links

Instructor / TAs	Office hours	Optional, ungraded discussions
Antonio de Leon	Fri 3–4 p.m. (in person) GSC Room 204	—
Jeannine Romero	Tue / Thu 10–11 a.m.	01A: Fri 9:20–10:25 a.m. R Carson Acad 240 (in person)
Ehsan Mirafzali	Mon 3–5 p.m.	01B: Fri 10:40–11:45 a.m. (Zoom)
Seokjun Choi	Wed 3–4 p.m. 5:05–6:05 p.m.	01C: Wed 4–5:05 p.m. (Zoom)

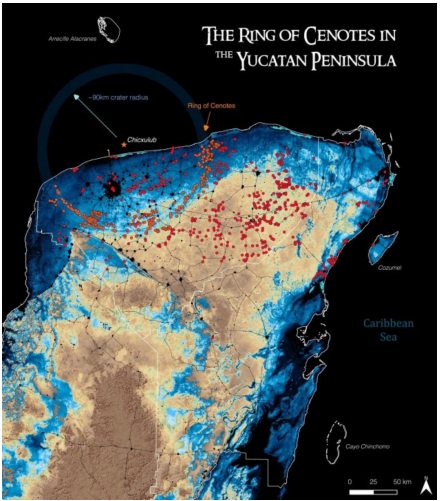
Upcoming deadlines HW and quizzes: **10 p.m. Pacific**

HW 3	**Tue, Oct 27**	Gradescope
Quiz 5	**Sun, Nov 1**	Canvas
HW 4	**Tue, Nov 3**	Gradescope

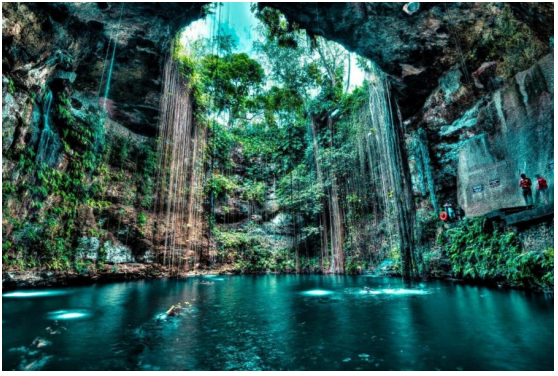
GSC: Graduate Student Commons. TA office-hour times link to Zoom.

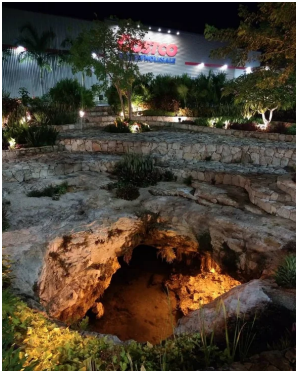
HW: 24-hour grace period. Quizzes: no grace period. Approved accommodations apply.



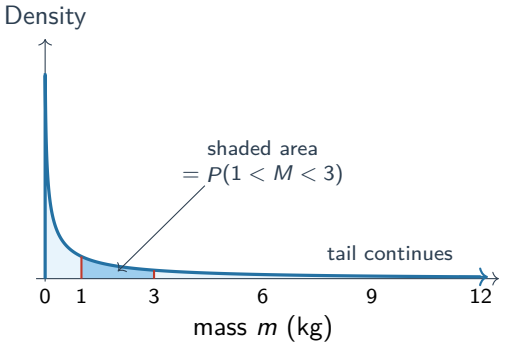

Arrecife Alacranes
THE RING OF CENOTES IN THE YUCATAN PENINSULA
~90km crater radius
Ring of Cenotes
Chicxulub
Cozumel
Caribbean Sea
Cayo Chinchorro
0 25 50 km







Let M be the total recovered mass from a meteorite fall, measured in kg.



A continuous measurement
Mass can take values throughout an interval. A density curve describes its probability distribution.

An interval probability
What is the probability that M lies between 1 and 3 kg?

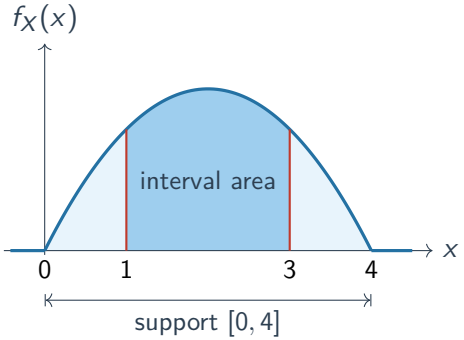
The shaded **area** gives that probability.

The curve is a **probability density**. Its total area, including the right tail, is 1.

Curve based on recorded meteorite masses. Data: NASA / The Meteoritical Society, *Meteorite Landings*.

Reading a probability density

A **probability density function (PDF)** f_X describes probabilities for a continuous random variable X through area under its curve.



Read the axes

x: a value of X.

$f_X(x)$: the density height there.

Identify the support

The support is $[0, 4]$: X takes values within this interval. The density is zero outside it.

Read the shading

The darker area is $P(1 < X < 3)$.

Probability density: definition and properties

The **support** S is the range of possible values of X. The density f_X is zero outside S.

Nonnegativity

$$f_X(x) \geq 0 \qquad \text{for all } x.$$

Density heights cannot be negative.

Total probability

$$\boxed{\int_S f_X(x)\,\mathrm{d}x = \int_{-\infty}^{\infty} f_X(x)\,\mathrm{d}x = 1.}$$

Integrate over the entire support.

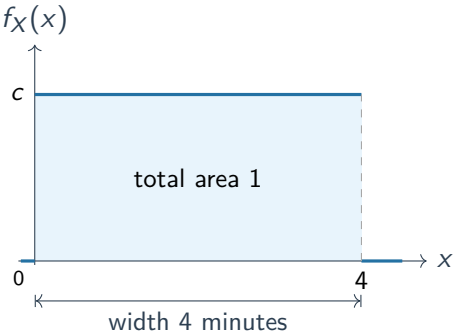
A **probability density function (PDF)** describes the distribution of X through interval probabilities:

$$\boxed{P(a < X < b) = \int_a^b f_X(x)\,\mathrm{d}x} \qquad (a < b).$$

For the preceding sketch, $S = [0, 4]$, so normalization requires $\int_0^4 f_X(x)\,\mathrm{d}x = 1$.

A constant density

Model a waiting time X with constant density c on $(0, 4)$ minutes, and zero density elsewhere.



The rectangle contains all probability, so its area is 1.

$$\underbrace{4}_{\text{width}} \; \underbrace{c}_{\text{height}} = 1.$$

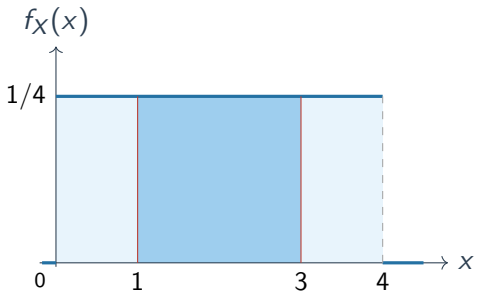
$$\boxed{c = \frac{1}{4}.}$$

The density has units of probability per minute.

A constant density on (a, b) is uniform. Its height is $1/(b-a)$, with zero density outside the interval.

Interval probability and point probability

For the same waiting-time model, $f_X(x) = 1/4$ on $(0, 4)$ and 0 elsewhere.



Find $P(1 < X < 3)$
Width $= 3 - 1 = 2$ minutes.
Area $= (1/4)(2) = 1/2$.

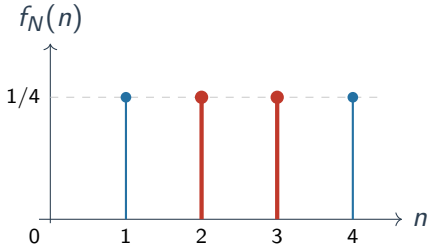
$$P(1 < X < 3) = \int_1^3 \frac{1}{4}\,\mathrm{d}u = \frac{1}{2}.$$

A width in minutes times a density per minute gives a probability.

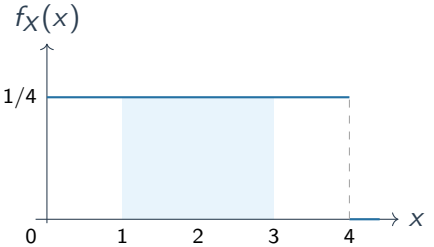
A single point has zero width: $P(X = 2) = 0$, even though $f_X(2) = 1/4$.
Including or excluding an interval's endpoints does not change its probability.

Probability mass and probability density

Let N have four equally likely values $1, 2, 3, 4$. Keep $X \sim \text{Unif}(0, 4)$ from the preceding example.



A PMF height is a point probability.



A PDF gives interval probability by area.

$$P(1 < N \leq 3) = \tfrac{1}{4} + \tfrac{1}{4} = \tfrac{1}{2}.$$

$$P(1 < X \leq 3) = \tfrac{1}{4}(3 - 1) = \tfrac{1}{2}.$$

For N, add point masses. For X, integrate the density over the interval.

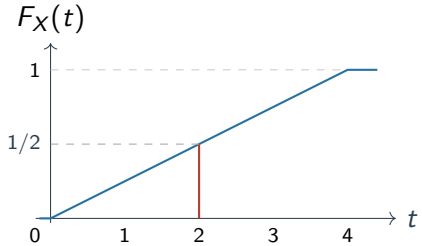
The symbol f is used for both functions. Here $P(N = 2) = 1/4$, while $P(X = 2) = 0$.

The CDF accumulates density area

$$F_X(t) = P(X \le t) = \int_{-\infty}^{t} f_X(u)\,\mathrm{d}u.$$

The same uniform model on $(0, 4)$

$$F_X(t) = \begin{cases} 0, & t \le 0, \\ \displaystyle\int_0^t \frac{1}{4}\,\mathrm{d}u = \frac{t}{4}, & 0 < t < 4, \\ 1, & t \ge 4. \end{cases}$$



$$F_X(2) = P(X \le 2) = 1/2.$$

$$P(1 < X \le 3) = F_X(3) - F_X(1) = \tfrac{3}{4} - \tfrac{1}{4} = \tfrac{1}{2}.$$

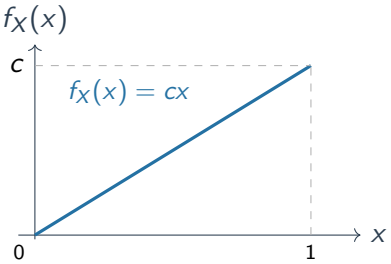
The cutoff t is fixed. The integration variable u varies up to it.

An increasing density

Consider $f_X(x) = cx$ on $(0, 1)$, and zero density elsewhere.

The triangle's area must be one

$$\frac{1}{2}\underbrace{(1)}_{\text{base}}\underbrace{c}_{\text{height}} = 1 \quad \Longrightarrow \quad c = 2.$$



Thus $f_X(x) = 2x$ for $0 < x < 1$.

Its integral confirms the area:

$$\int_0^1 2x \, \mathrm{d}x = [x^2]_0^1 = 1.$$

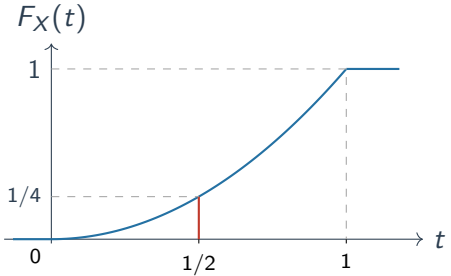
Equal-length intervals farther to the right receive more probability because the density is higher there.

The CDF of an increasing density

Let $f_X(x) = 2x$ for $0 < x < 1$, and zero density elsewhere.

$$F_X(t) = \begin{cases} 0, & t \le 0, \\ \displaystyle\int_0^t 2u \, \mathrm{d}u = t^2, & 0 < t < 1, \\ 1, & t \ge 1. \end{cases}$$

Find $P(X > 1/2)$.



The CDF height $F_X(1/2) = 1/4$ equals the density area from 0 to $1/2$.

The interior formula t^2 applies only for $0 < t < 1$. Outside the support, the CDF is 0 or 1.

Density, cumulative probability, and a point

Inside $(0, 1)$, the CDF is $F_X(t) = t^2$. Its slope recovers the density:

$$F_X'(t) = 2t = f_X(t).$$

Outside this interval, the CDF is flat and the density is zero.

A density height can exceed 1. Probabilities remain between 0 and 1.

Quantity	Value	Meaning
$f_X(3/4)$	$3/2$	density height
$F_X(3/4)$	$9/16$	probability $X \leq 3/4$
$P(X = 3/4)$	0	point probability

Integrating the PDF gives the CDF. Differentiating the CDF gives the PDF where the derivative exists.

Changing density values at isolated endpoints does not change any probability.

Expectation as a weighted average

Expectation averages values using their distribution. For a continuous variable, use an integral with density weights.

Discrete X

$$\mathbb{E}[X] = \sum_{x \text{ in the support}} x\, f_X(x).$$

The PMF gives a probability weight at each possible value.

Continuous X with support S

$$\boxed{\mathbb{E}[X] = \int_S x\, f_X(x)\, \mathrm{d}x.}$$

Multiply each value by the density and integrate over the support.

For a function of X, average the function using the same density:

$$\mathbb{E}[g(X)] = \int_S g(x) f_X(x)\, \mathrm{d}x.$$

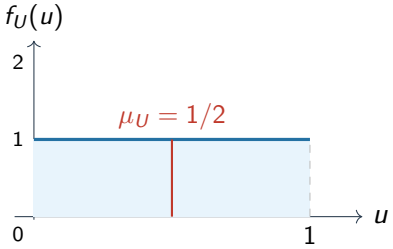
Use $g(x) = x$ for the mean and $g(x) = x^2$ for the second moment.

The integration variable runs through S. The result is a number. A finite expectation requires $\int_S |g(x)| f_X(x)\, \mathrm{d}x < \infty$.

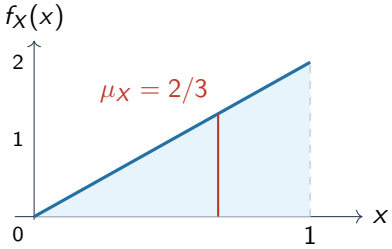
Means under two densities

Compare a uniform density with the increasing density on the **same interval** $(0, 1)$.

Uniform: $f_U(u) = 1$



Increasing: $f_X(x) = 2x$



$$\mathbb{E}[U] = \int_0^1 u(1)\,\mathrm{d}u = \frac{1}{2}.$$

$$\mathbb{E}[X] = \int_0^1 x(2x)\,\mathrm{d}x = \frac{2}{3}.$$

The increasing density gives greater weight to larger values, shifting its mean to the right.

Second moments and variance

Variance is the second moment minus the square of the mean.

Uniform on $(0,1)$

$$\mathbb{E}[U^2] = \int_0^1 u^2(1)\,\mathrm{d}u = \frac{1}{3}.$$

$$\mathrm{Var}(U) = \frac{1}{3} - \left(\frac{1}{2}\right)^2 = \frac{1}{12}.$$

Increasing density $2x$

$$\mathbb{E}[X^2] = \int_0^1 x^2(2x)\,\mathrm{d}x = \frac{1}{2}.$$

$$\mathrm{Var}(X) = \frac{1}{2} - \left(\frac{2}{3}\right)^2 = \frac{1}{18}.$$

More generally, for $V \sim \mathrm{Unif}(a,b)$,

$$\boxed{\mathbb{E}[V] = \frac{a+b}{2}, \qquad \mathrm{Var}(V) = \frac{(b-a)^2}{12}.}$$

These formulas follow from $V = a + (b-a)U$ and the affine rules from Week 4.

Practice: a decreasing density

Suppose $f_W(w) = c(1 - w)$ for $0 < w < 1$, and zero elsewhere.

1. Find c and give $F_W(t)$ for every real t.
2. Predict whether $\mathbb{E}[W]$ is below or above $1/2$, then compute it.

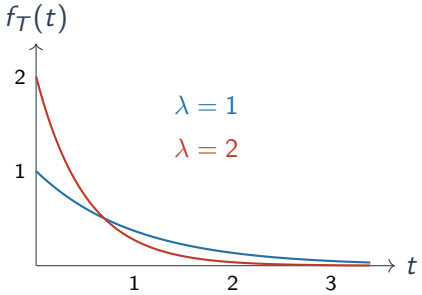
For review: HW4 Q1–3. Uniform formulas from today also support Q4(a).

An exponential waiting time

Assume a positive waiting time $T \sim \text{Exp}(\lambda)$, with rate $\lambda > 0$.

$$f_T(t) = \lambda e^{-\lambda t} I_{\{t>0\}}$$

Normalization



Time in minutes; rates per minute.

A larger rate places more probability near zero. Each curve has total area one.

The exponential law is a modeling assumption. Waiting times need not follow it.

CDF and survival function

As on Monday, integrate the density up to a fixed cutoff $t \geq 0$. Here u varies from 0 to t.

$$F_T(t) = \int_0^t \lambda e^{-\lambda u} \, \mathrm{d}u.$$

The survival function gives the probability that the wait exceeds t.
At $t = 0$, $F_T(0) = 0$ and $P(T > 0) = 1$.

Exponential moments and rate units

If $T \sim \text{Exp}(\lambda)$, then

$$\mathbb{E}[T] = \int_0^\infty t\lambda e^{-\lambda t}\,\mathrm{d}t = \frac{1}{\lambda},\ \mathbb{E}[T^2] = \frac{2}{\lambda^2}.$$

$$\text{Var}(T) = \frac{1}{\lambda^2},\ \text{SD}(T) = \frac{1}{\lambda}.$$

Example. A processing time has rate 2 per minute. Find its mean, variance, and $P(T > 0.5)$.

Rate has units of inverse time. The reciprocal rate is the mean waiting time.

The moment derivations by integration by parts appear in the optional appendix.

The remaining wait

Let $T \sim \text{Exp}(\lambda)$. For $s, t \geq 0$, suppose the wait has already exceeded s.



Derive $P(T > s + t \mid T > s)$ using event conditioning.

Under this model, elapsed time does not change the remaining waiting-time law.

The property is called **memorylessness**: t is an additional wait, while $s + t$ is a total wait.

A uniform wait has a different remaining wait

Let $U \sim \text{Unif}(0, 10)$, in minutes. The original wait is at most ten minutes. **Compare** $P(U > 7 \mid U > 4)$ with $P(U > 3)$.

Memorylessness is a property of the exponential model. A positive waiting time need not have that property.

The normal density

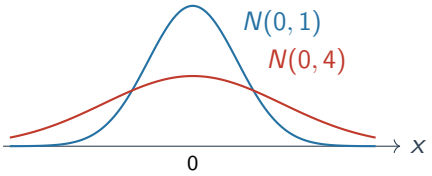
A normal model describes measurements that fluctuate symmetrically around a center. For $\sigma > 0$,

$$X \sim N(\mu, \sigma^2), \; f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}, \; x \in \mathbb{R}.$$

The standard normal model

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$

$$\Phi(z) = P(Z \le z) = \int_{-\infty}^{z} \phi(u)\,\mathrm{d}u$$



Same mean, different standard deviations.

Here $Z \sim N(0,1)$. The lower-case ϕ is a density and upper-case Φ is its CDF.

μ is the mean. σ^2 is the variance. Increasing σ spreads out the density while preserving total area one.

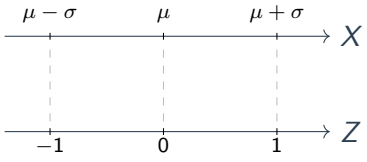
Standardization in a normal model

If $X \sim N(\mu, \sigma^2)$ with $\sigma > 0$, then

$$\boxed{Z = \frac{X - \mu}{\sigma} \sim N(0, 1).}$$

The event changes units.

$$X \leq b \iff Z \leq \frac{b - \mu}{\sigma}.$$



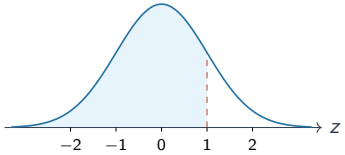
Subtract μ to center the scale. Divide by the SD σ to measure distance in standard deviations.

In $N(\mu, \sigma^2)$, the second parameter is the variance. Standardization uses its square root.

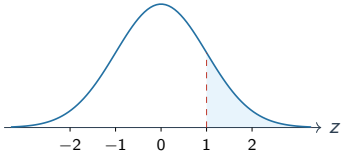
This gives a standard normal law because X is normal. Standardizing an arbitrary distribution does not make it normal.

Left, right, and interval probabilities

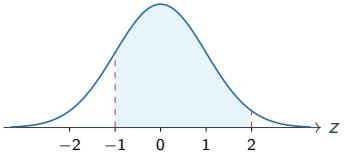
For $Z \sim N(0,1)$, $\Phi(z)$ is the area to the left of z.



$P(Z \le 1) = \Phi(1)$



$P(Z > 1) = 1 - \Phi(1)$



$P(-1 < Z < 2)$
$= \Phi(2) - \Phi(-1)$

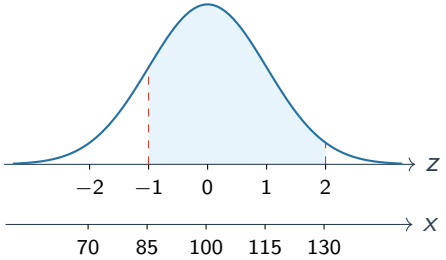
Use symmetry to express $\Phi(-z)$ through $\Phi(z)$.

z	0	1	2
$\Phi(z)$	0.500000000	0.841344746	0.977249868

The table gives numerical approximations. Shading extends into the unbounded tails.

A measurement between two limits

Assume a measurement $X \sim N(100, 225)$. Find $P(85 < X < 130)$.



First identify SD: $\sigma = \sqrt{225}$.

Original values align with standardized values.

Transform both endpoints. Centering and division by SD preserve their order.

Practice: normal tail probabilities

Assume $W \sim N(80, 16)$. Use $\Phi(1) = 0.841345$ and $\Phi(2) = 0.977250$.

1. Find $P(W > 84)$.
2. Find $P(|W - 80| > 8)$.

Parameters and probabilities

	$\text{Unif}(a, b)$	$\text{Exp}(\lambda)$	$N(\mu, \sigma^2)$
Range	(a, b)	$(0, \infty)$	$\mathbb{R}$
Mean	$(a + b)/2$	$1/\lambda$	μ
Variance	$(b - a)^2/12$	$1/\lambda^2$	σ^2
$F_X(t)$	$(t - a)/(b - a)$	$1 - e^{-\lambda t}$	$\Phi((t - \mu)/\sigma)$

λ is a rate. σ^2 is a variance. Standardization uses σ.

The uniform CDF formula applies for $a < t < b$ (outside: 0 or 1). The exponential formula applies for $t \geq 0$ (otherwise: 0).

Week 4 described a pair (X, Y) by the mass at each possible combination. We now use a joint density over the plane.

Discrete joint PMF

$$f_{X,Y}(x, y) = P(X = x, Y = y).$$

For an event region A, add the masses of the points in A.

Continuous joint PDF

$$P((X, Y) \in A) = \iint_A f_{X,Y}(x, y)\,\mathrm{d}x\,\mathrm{d}y.$$

Integrate the density over the part of A inside the support.

A joint density is nonnegative and has total integral one:

$$f_{X,Y}(x, y) \geq 0, \quad \iint_{\mathbb{R}^2} f_{X,Y}(x, y)\,\mathrm{d}x\,\mathrm{d}y = 1.$$

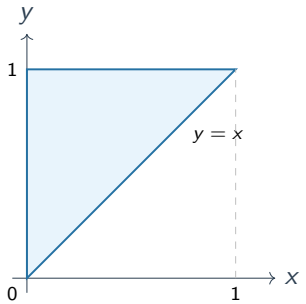
The joint PDF is zero outside its support. The region determines the limits, and the density supplies the probability weights.

The support of an ordered pair

Let X be a normalized start time and Y the finish time. Assume constant density on

$$S = \{(x, y) : 0 < x < y < 1\},$$

and zero elsewhere.



The inequalities describe the region together:

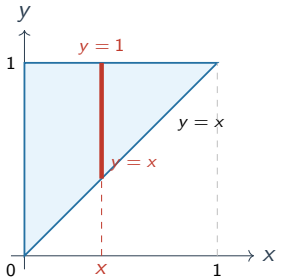
- $x > 0$: to the right of the vertical axis.
- $y > x$: above the diagonal $y = x$.
- $y < 1$: below the horizontal line $y = 1$.

The enclosing square contains pairs that the model never allows.

Uniformity over the triangle is the model assumption.

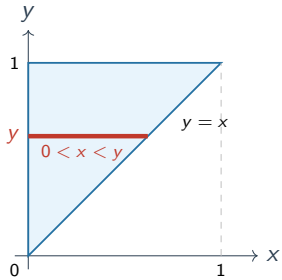
Reading the integration limits

The triangle $0 < x < y < 1$ can be described in either order.



Inner integral: fix x and vary y.

$$\int_0^1 \underbrace{\left[\int_x^1 f_{X,Y}(x,y)\,\mathrm{d}y\right]}_{\text{a function of } x} \mathrm{d}x$$



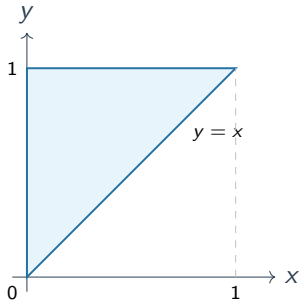
Inner integral: fix y and vary x.

$$\int_0^1 \underbrace{\left[\int_0^y f_{X,Y}(x,y)\,\mathrm{d}x\right]}_{\text{a function of } y} \mathrm{d}y$$

The inner integral returns a function. The outer integral then integrates that function.

A uniform law on a triangle

A normalized start/finish pair has constant joint density on $S = \{(x, y) : 0 < x < y < 1\}$.



$$f_{X,Y}(x, y) = c I_{\{0<x<y<1\}}$$

Use constant height $\times$ support area:

$$1 = c\left(\frac{1}{2}\right), \qquad c = 2.$$

Check with an iterated integral.

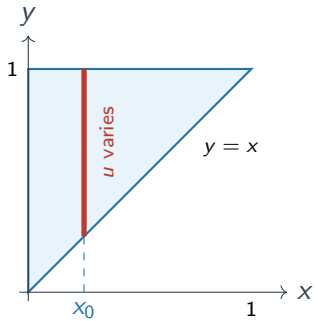
Uniformity over the triangle is the model assumption.

Uniformity over the triangle does not make either marginal distribution uniform.

A marginal density keeps one coordinate

To find $f_X(x_0)$, hold x_0 fixed and integrate over the Y coordinate.

$$f_X(x_0) = \int_{-\infty}^{\infty} f_{X,Y}(x_0, u)\,\mathrm{d}u.$$



Fixed: the argument x_0.
Varying: u, in the Y coordinate.

In the triangular model, set $x_0 = 1/4$:

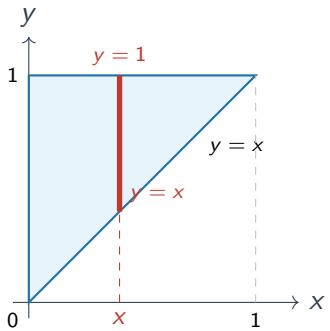
$$f_X(1/4) = \int_{1/4}^{1} 2\,\mathrm{d}u = \frac{3}{2}.$$

The result is a density height. It can exceed one.

Fixing the argument evaluates a density. It does not condition on an observation.

The marginal density of the start

For $f_{X,Y}(x,y) = 2I_{\{0<x<y<1\}}$, fix x and integrate over every compatible y.

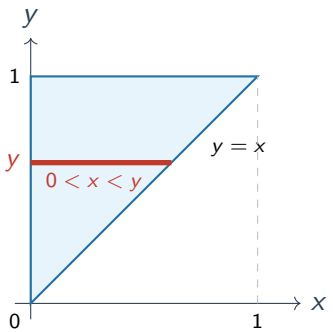


The projection onto the x-axis is $(0,1)$.

Find $f_X(x)$, including outside the projection.

The marginal density of the finish

Fix y. The compatible start coordinates satisfy $0 < x < y$.

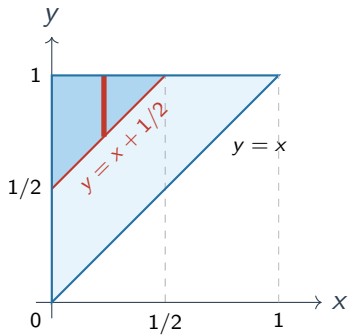


Find $f_Y(y)$ and check its integral.

The two marginals describe different coordinates of the same joint model.

A duration longer than one half

Find $P(Y - X > 1/2)$. Intersect the duration condition with $0 < x < y < 1$.

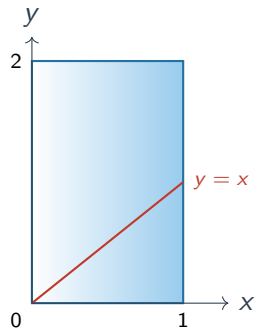


Read the limits from the shaded region.

Other order: $\int_{1/2}^{1} \int_{0}^{y-1/2} 2 \, \mathrm{d}x \, \mathrm{d}y$. Both integrals cover the same region.

A nonuniform joint density

Practice. Let $f_{X,Y}(x,y) = x I_{\{0<x<1,\ 0<y<2\}}$.



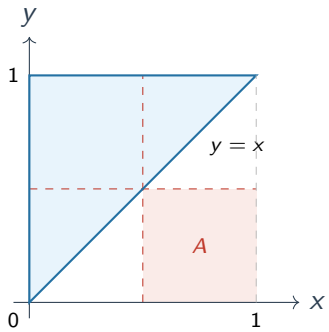
Density increases continuously with x.

The model is normalized: $\int_0^1 \int_0^2 x \, \mathrm{d}y \, \mathrm{d}x = 1$.

Shade $Y < X$, then calculate its probability.

Dependence from impossible combinations

Return to the triangular model. Consider $A = \{X > 1/2,\ Y < 1/2\}$.



Compare the joint-event probability with the product of marginal probabilities.

Independence would require these event probabilities to factor.

Independent continuous random variables

Independence means that, for every pair of sets A and B,

$$P(X \in A, Y \in B) = P(X \in A)\, P(Y \in B),$$

or equivalently $P(X \in A \mid Y \in B) = P(X \in A)$ whenever $P(Y \in B) > 0$. For variables with a joint density, independence holds exactly when a joint PDF can be written as

$$\boxed{f_{X,Y}(x, y) = f_X(x) f_Y(y)}.$$

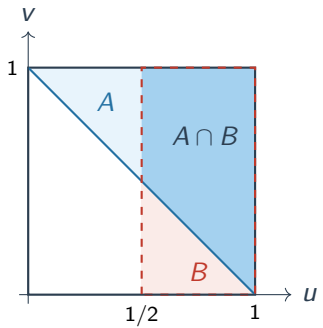
Example. Let U and V be independent $\mathrm{Unif}(0, 1)$ variables. Then

$$f_{U,V}(u, v) = \begin{cases} 1, & 0 < u < 1,\ 0 < v < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Independence lets us build the joint density from the two marginal densities. The joint support must also agree.

An event condition within a joint model

Let U, V be independent Unif$(0, 1)$ variables. Define the events $A = \{U + V > 1\}$ and $B = \{U > 1/2\}$.



Find $P(A)$ and $P(A \mid B)$.

Dashed rectangle: B.

The sum uses U. Learning that $U > 1/2$ increases the chance that the sum exceeds one, even though U and V are independent.

Reading selections

Harchol-Balter, supplied 2024 version

Monday	pp. 134–137 and 141–142: density/CDF, uniform model, moments.
Wednesday	pp. 138–139, 142 (exponential); pp. 170–175 (normal moments and probabilities).
Friday	pp. 153–155: joint densities and marginals. Stop before conditional densities.

DeGroot–Schervish, fourth edition

Densities and moments	pp. 101–104, 111–112, 209–210; continuous expectation statement on pp. 213–214.
Named models	p. 303 and pp. 307–308 (normal); pp. 321–322, exponential portions only.
Joint laws	pp. 120–122 and the marginalization theorem on p. 132; p. 133 for the optional curved example.

Printed pages. Read selected passages alongside lecture; whole chapters are not assigned.

Exclude surrounding MGF, gamma, conditional-density, normal-sum, and CLT material. The appendix contains optional derivations and extensions.

Exponential mean and variance

Optional derivation or extension.

Use integration by parts on $[0, b]$, then let $b \to \infty$.

$$\mathbb{E}[T] = \int_0^\infty t\lambda e^{-\lambda t}\,\mathrm{d}t,\ \mathbb{E}[T^2] = \int_0^\infty t^2\lambda e^{-\lambda t}\,\mathrm{d}t.$$

The boundary terms vanish: $b^k e^{-\lambda b} \to 0$ for $k = 1, 2$.

Standard-normal moments

Optional derivation or extension.

Let Z have density ϕ. Its symmetry and decaying tails give $\mathbb{E}[Z] = 0$.

$$\phi'(z) = -z\phi(z),\ \mathbb{E}[Z^2] = \int_{-\infty}^{\infty} z^2\phi(z)\,\mathrm{d}z.$$

Integrate by parts using $z^2\phi(z) = -z\phi'(z)$.

Use finite bounds first. $z\phi(z) \to 0$ at both ends, and $\int |z|\phi(z)\,\mathrm{d}z < \infty$.

Why standardization works

Optional derivation or extension.

Let $X \sim N(\mu, \sigma^2)$ and set $Z = (X - \mu)/\sigma$, where $\sigma > 0$. **Find the CDF of Z.** Use $u = (x - \mu)/\sigma$ and $\mathrm{d}x = \sigma \, \mathrm{d}u$.

Thus $Z \sim N(0, 1)$. Since $X = \mu + \sigma Z$,

The CDF establishes the normal law. Mean zero and variance one alone do not.

Gaussian normalization

Optional derivation or extension.

Optional proof. Requires double integration and the polar area element.

Put $I = \int_{\mathbb{R}} e^{-z^2/2}\,\mathrm{d}z$. We seek the normalizing constant.

$$I^2 = \iint_{\mathbb{R}^2} e^{-(x^2+y^2)/2}\,\mathrm{d}x\,\mathrm{d}y.$$

Use $x = r\cos\theta$, $y = r\sin\theta$, and area element $r\,\mathrm{d}r\,\mathrm{d}\theta$.

$I < \infty$: for $|z| \geq 1$, $e^{-z^2/2} \leq e^{-|z|/2}$. Polar area is radius $\times$ radial width $\times$ angular width.

A normal percentile

Optional derivation or extension.

Optional application. A percentile is a threshold in a specified distribution.

For $X \sim N(\mu, \sigma^2)$ and $0 < p < 1$, find q_p such that $P(X \leq q_p) = p$.

For $N(100, 225)$, use $\Phi^{-1}(0.975) \approx 1.96$.

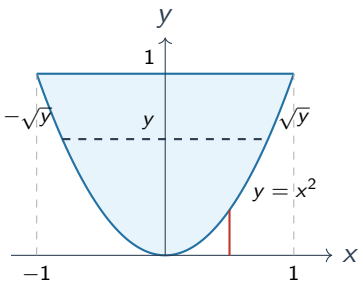
About 97.5% of this model's probability lies below the threshold.

A density on curved support

Optional derivation or extension.

Optional extension. The same slicing method applies to a curved boundary.

Let $f_{X,Y}(x,y) = cx^2 y I_{\{x^2<y<1\}}$.



Find c. The x-projection is $(-1, 1)$.

Marginals on curved support

Optional derivation or extension.

Optional extension. Here $f_{X,Y}(x,y) = (21/4)x^2 y I_{\{x^2<y<1\}}$.

Vertical slice

$-1 < x < 1$, $x^2 < y < 1$.

Horizontal slice

$0 < y < 1$, $-\sqrt{y} < x < \sqrt{y}$.

The horizontal slice uses both branches, $-\sqrt{y}$ and $\sqrt{y}$.